

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

Deemed to be University, Bhubaneswar - 751024, Odisha

MID SEMESTER EXAMINATION — EXAMINATION QUESTION PAPER

Subject: Differential Equation & Linear Algebra

Time: 1.5 Hours

Semester / Programme: First Semester

Full Marks: 20

Instructions: Answer Question No. 1 (Compulsory) and any THREE from Question No. 2 to 5. Figures in brackets indicate marks.

SECTION - A (Compulsory)

1(a) Find the integrating factor of the first-order linear differential equation $x \frac{dy}{dx} + yx = 1$. [1]

$$I.F. = e^{\int P(x)dx}$$

1(b) Determine the Wronskian $W(y_1, y_2)$ of the two solutions $y_1 = e^{2x}$ and $y_2 = e^{-3x}$ of a second-order linear differential equation. [1]

$$\text{Formula: } W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

1(c) Evaluate the improper integral $\int_1^\infty (1/x^2) dx$. [1]

$$\int_a^\infty f(x)dx = \lim_{t \rightarrow \infty} \int_a^t f(x)dx$$

1(d) State the relationship between the Gamma function $\Gamma(n)$ and the Beta function $B(m, n)$ for positive real numbers m and n . [1]

$$B(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$$

1(e) Find the general form of the complementary function (C.F.) for the second-order differential equation $(D^2 - 4D + 4)y = 0$. [1]

$$m^2 - 4m + 4 = 0$$

SECTION - B (Answer any THREE)

Answer any THREE questions from Q.2 to Q.5. Each question carries 5 marks.

Q.2. (a) Solve the exact differential equation: $(2x + y^3)dx + (3xy^2 + 4)dy = 0$. [2.5]

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

(b) Find the orthogonal trajectories of the family of curves given by $y = Cx^2$, where C is an arbitrary constant. [2.5]

$$\frac{dy}{dx} \rightarrow -\frac{dx}{dy}$$

Q.3. (a) Solve the Bernoulli differential equation: $(dy/dx) + (y/x) = x^2 y^2$. [2.5]

$$v = y^{1-n}$$

(b) A thermometer reading 70°F is placed in an oven preheated to a constant temperature of 300°F . Through Newton's Law of Cooling/Heating, if the thermometer reads 150°F after 2 minutes, determine the temperature at $t = 5$ minutes. [2.5]

$$\frac{dT}{dt} = -k(T - T_m)$$

Q.4. (a) Solve the Euler-Cauchy equation: $x^2 y'' - 3xy' + 4y = 0$. [2.5]

$$m(m-1) + am + b = 0$$

(b) Find the particular integral (P.I.) of the differential equation $(D^2 + 4)y = (2x)$ using operator methods. [2.5]

$$P.I. = \frac{1}{f(D)}Q(x)$$

Q.5. (a) Solve the nonhomogeneous differential equation $y'' - 2y' + y = (e^x / x)$ using the method of variation of parameters. [2.5]

$$P.I. = -y_1 \int \frac{y_2 Q}{W} dx + y_2 \int \frac{y_1 Q}{W} dx$$

(b) Test the convergence of the improper integral $\int_0^1 (1/x^{1/2}) dx$. [2.5]

$$\int_0^a \frac{1}{x^p} dx \text{ converges if } p < 1$$

--- END OF PAPER ---